Daily Drilling Report
SPD19B-03
8/20/2014

PETROPARS Oil & Gas Developer

N.I.O.C. Pars Oil & Gas Company

Well Data

Well Name	Operator	Field Name	Well Type 1	Rig	Code
SPD19B-03	Petropars Ltd.	South Pars Gas Field	Development	Deep Driller, DD6	SP19-3B-01-DL-FOR-0...
Report Number	**Hole Section**	**Days LTI (days)**	**Water Depth (m)**	**Original KB Elevation (m)**	**Head Count**
24.0	12 1/4"	169.00	72.83	31.70	97.0
Last Cas	**Daily Cost Total**	**Daily Mud Cost**	**Cum Cost To Date**	**AFE Number**	**Cum Time Log Days (days)**
13 3/8" Shoe, 1,487.00mKB	44,333.81	14,244.50	5,483,529.66		20.85
Depth Start (mKB)	**Depth End (mKB)**	**Depth End (TVD) (mKB)**	**Depth Progress (m)**	**Drilling Hours (hrs)**	**Average ROP (m/hr)**
1,492.00	1,492.00	0.00			

Operations at Report Time
Drilling new formation @ 1492m.

Operations Summary
Skid f/ well B-05 to B-03. N/U 18 3/4'' BOP. L/D old directional BHA. Perform BOP test. RIH rotary BHA#7 to 1100m. Slip & Cut, in progress.

Operations next 24hrs
Cont' Slip & Cut. Resume RIH rotary BHA and D/O Shoe track. Drill 3m new formation. Perform FIT. POOH to surface while L/D high rotating hours DP.

Supervisors Contact

Job Contact	Position
Mojtaba Parvaresh	Drilling Supt.
Mohammadreza Asaie	Senior Drilling Sup.
Jabbar Mohammadi	Night Drilling Sup.
Hamed Daneshkhah / B.Khalilmoghadam	Drilling Engineer
Ali Sahebnouri	Geologist
Soroush Tehrani	Completion Eng.
R. Kamalpour	POGC Rep.

HSE Data

Type	Last Date	Next Date
Abandon Drill	8/10/2014	9/30/2014
BOP Test	8/20/2014	9/11/2014
Chemical Spill Drill	8/17/2014	10/17/2014
Fire Drill	8/10/2014	9/30/2014
H2S Drill	8/16/2014	9/16/2014
Heli Crash Drill	7/15/2014	9/14/2014
Medical Drill	7/26/2014	9/25/2014
MOB Drill	4/16/2014	6/16/2014
Pit Drill	8/2/2014	9/17/2014
Trip Drill	8/3/2014	9/23/2014

STOP Cards

Number of Reports: 30

Operations Summary

Start Time	End Time	Dur (hrs)	Phase	Code 1	Code 2	Time P-N-T	Comment
05:00	06:00	1.00	12 1/4"	B	01	P	HPJSM. Prepare and skid f/ well B-05 to B-03.
06:00	08:30	2.50	12 1/4"	C	05	P	N/U 18 3/4" BOP stack and tighten drive lock bolts, N/U Mandrel + Overshot + diverter and mouse hole.
08:30	09:00	0.50	12 1/4"	C	05	P	P/U & M/U Aker long wear bushing + RT to one std 5 7/8" DP. Set WB inside WH. POOH and L/D RT.
09:00	11:00	2.00	12 1/4"	D	01	P	P/U & RIH old directional BHA#11 from derrick and L/D 7'' TM Sub + 8 1/2'' Float sub + 2 jts 8'' NMDC + Restrictor Sub + 12'' string stabilizer + 8'' Pony NMDC + 9 5/8'' Motor.
11:00	11:30	0.50	12 1/4"	C	05	P	P/U and M/U 13 5/8" wear bushing RT to one std 5 7/8" DP, RIH and retrieve 13 5/8" WB.
11:30	15:30	4.00	12 1/4"	C	12	P	P/U and M/U Aker 13 5/8" BOP test plug to 2 jts of 5 7/8" DP's + FOSV+ 1 x 5 7/8" pup jt. RIH and install test plug inside well head at 18.94 m. L/U to CMT unit. Flush lines and surface line pressure test to 500/5000 psi for 5/10 min. Fill up BOP stack. Press test 18 3/4" BOP stack + 5 7/8" Lower pipe rams + (3 1/2" x 7 5/8") Variable pipe rams + 5 7/8" Upper pipe rams + Annular BOP + manual choke and kill valves + HCRs + FOSV to 500/5000 psi for 5/10 min, Ok. Back off test plug landing string and pressure test Blind/Shear rams + Kill & choke hoses to 500/5000 psi for 5/10 min, Ok. ** While fill up BOP Stack, observe leaking from side out let. POOH, found Dovetail packing of BOP test plug damaged. Change out same and RIH again, ok.
15:30	16:00	0.50	12 1/4"	C	01	P	Screw into test plug and retrieve same. POOH and L/D 5 7/8" DP's + Test Assy.
16:00	16:30	0.50	12 1/4"	C	05	P	P/U & M/U Aker long wear bushing + RT to one std 5 7/8" DP. Set WB inside WH. POOH and L/D RT.
16:30	17:30	1.00	12 1/4"	D	04	P	P/U and M/U rotary BHA#07 consists of: 12 1/4" Insert bit-RR (CH04JMRSV, IADC: 417, SN: 255776, Nozzles: 3 x 18/32" + 1 x 16/32" CJ) + Bit sub with float valve inside + 12 1/8" Stabilizer (COR3057AD) + 3 jts 8 1/4" DC + X/O + 15 jts 5 7/8" HWDP. RIH to 172m.
17:30	23:00	5.50	12 1/4"	D	04	P	RIH rotary BHA#07 on 5 7/8" DP f/ 172m to 1100m. ** P/U 39 jts of 5 7/8" New DP joint by joint from cantilever in RIH BHA. Flush string. >>> Mix and prepare 1600 bbl 9.1 ppg Polymer Mud.
23:00	00:00	1.00	12 1/4"	D	20	P	Perform Slip & Cut, in progress.

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Operations Summary

Start Time	End Time	Dur (hrs)	Phase	Code 1	Code 2	Time P-N-T	Comment
00:00	00:00		12 1/4"				Aug 21, 2014 00:00 - 01:00 Cont' Slip & Cut 110 ft drill line. 01:00 - 03:00 Cont' RIH rotary BHA on 5 7/8" DP f/ 1100m to 1435m. Wash down to 1461.8m. Tag CMT Plug w/ 10 kips. 03:00 - 06:00 D/O cement plugs @ 1461.8 m + Float Collar @ 1462.8m + cement inside shoe track + Float Shoe @ 1486.7m and clear rathole to 1492m w/ 300-900 gpm/ 200-1100 psi/ 5-30 kips/ 50 rpm/ 9-16 kft-lbf. * Pump 30 bbls Hi-Vis pill after drill out Float Collar and middle of shoe track. * Reciprocate string inside float collar and shoe with and without rotation, Ok. * Start displacing at 1483m: Pump 30 bbls Hi-Vis + 50 bbls SW and displace hole to 9.1 ppg Polymer Mud.

Mud Information

Depth (mKB)	Type	Density (lb/gal)	Funnel Viscosity (s/qt)	PV Calc (cp)	YP Calc (lbf/100ft²)	Filtrate (mL/30min)
pH	Pf (mL/mL)	Mf (mL/mL)	Pm (mL/mL)	Sand (%)	Low Gravity Solids (%)	High Gravity Solids (%)
Vis 3rpm	Vis 6rpm	Percent Oil (%)	Percent Water (%)	Solids (%)	Chlorides (mg/L)	Filter Cake (/32")
Gel 10 sec (lbf/100ft²)	Gel 10 min (lbf/100ft²)	Gel 30 min (lbf/100ft²)	Oil Water Ratio	T (fl) (°C)	Active Mud Volume (bbl)	Reserve Mud Volume (bbl)
Cum Mud Added (bbl)	Mud Lost to Surface (bbl)	Mud Lost to Hole (bbl)	Avg Background Gas (%)	Avg Drill Gas (%)	K	n

Marine Conditions

Swell Height (m)	Visibility (km)	Wind Direction (°)	Wind Speed (knots)	Barometric Pressure (mmHg)	Temperature - High (°C)
0.50	7.	245	10.0	998.0	35.0

Support Vessels

Vessel Name	Type	Arrival Date	Depart	Note
Mataf Pride	Crew Boat	8/20/2014 22:15	8/20/2014 23:25	Transfrd 27 pxs from Kish & 34 pxs for Kish f/ DD6.
Marina-1	Supply Boat	8/16/2014 15:15		Unable to release Rig's mooring rope f/ 2300 Hrs due to critical twist on bollard boat. WOW.
Borkan	Supply Boat	8/19/2014 21:00		B/L Cargo for SHJ / Kish, C/Off & Dep for Kish. ETA Kish @ 0800 Hrs 21-08-14.
Morshed	Supply Boat	8/19/2014 09:05		A/S DD6 embarked 2 pax for Aban8, C/Off & Dep for Sahar2. ETA-45Min.
Ahmed VI	Supply Boat	8/19/2014 09:40		A/S DD6 Transfrd 3 pxs from Sahar 1, C/Off & Dep @ 1710 Hrs for Sahar 2.

Mud Additives

Description	Consumed	Cum Rec	On Loc
MI-Cide		15.0	15.0
Spersene		145.0	145.0
Calcium Carbonate Coarse		3.0	3.0
CaCo3-F BB		9.0	9.0
Alpine Drill Beads Coarse		72.0	72.0
Pipe-Lax		6.0	6.0
Mica Coarse		80.0	80.0
Safe - Scav		32.0	32.0
Mica Medium		80.0	80.0
Magcolube		24.0	24.0
Quick Seal Medium		192.0	192.0
Conqor 202		4.0	4.0
Barite Bulk PDF		0.0	0.0
Mica Fine		160.0	160.0
Sodium Bicarbonate		110.0	110.0
CaCo3-M BB		4.0	4.0
Drilling Detergent		8.0	8.0
KCL		17.0	17.0
Nut Plug Medium		60.0	60.0
Nut Plug Fine		90.0	90.0
Quick Seal Coarse		96.0	96.0
Quick Seal Fine		64.0	64.0
PTS (Temp. Stabilizer)		3.0	3.0
Barite Bulk		179.0	179.0
Alpine Drill Beads Fine		58.0	58.0
CaCo3-C 25Kg		40.0	40.0
Pac-R		39.0	39.0
Citric Acid BG		150.0	150.0
Glydril		12.0	12.0
Defoamer		8.0	8.0
Duo-Tec	16.0	115.0	99.0

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Mud Additives

Description	Consumed	Cum Rec	On Loc
Caustic Soda 2014	20.0	148.0	128.0
Soda Ash 25 Kg	60.0	150.0	90.0
Pac-LV	65.0	262.0	197.0
Starch	140.0	280.0	140.0

Bulk Amounts

Supply Item Description	Consumed	Received	Cum On Loc
Barite Bulk	0.0	249.0	249.0
Cement G	0.0	57.0	57.0
Bentonite Bulk	0.0	8.0	8.0
Diesel Fuel	5.0	152.0	147.0
Potable Water	20.0	136.0	116.0
Drill Water	24.0	365.0	341.0
Barite BB	0.0	0.0	0.0
Bentonite BB	0.0	0.0	0.0

Bit/BHA Information

BHA No.	Size (in)	Make	Model	Serial Number	Bit Type	IADC Codes	Nozzles (/32")
7	12 1/4	Varel	CH04JMRSV	255776	Insert	417	18/18/18/16
TFA (in²)	**Drilling Time (hrs)**	**Bit Revs**	**Depth In (mKB)**	**Depth Out (mKB)**	**Depth Drilled (m)**	**BHA ROP (m/hr)**	**IADC Bit Dull**
0.94	0.00						-------
Flow Rate (min) (gpm)	**Flow Rate (max) (gpm)**	**SPP (min) (psi)**	**SPP (max) (psi)**	**WOB (min) (1000lbf)**	**WOB (max) (1000lbf)**	**RPM (min) (rpm)**	**RPM (max) (rpm)**
Circulating Time (hrs)	**Rotating Time (hrs)**	**% Rotating Time (%)**	**% Sliding Time (%)**	**PU HL (max) (1000lbf)**	**Rot HL (max) (1000lbf)**	**SO HL (min) (1000lbf)**	**String Wt (1000lbf)**

Drilling Parameters

Depth Start (mKB)	Depth End (mKB)	Drilling Time (hrs)	Drilling Torque	Flow Rate (gpm)	RPM (rpm)	SPP (psi)	WOB (1000lbf)	Drill Str Wt (1000lbf)

Drill String Components

Item Description	OD (in)	ID (in)	Len (m)	Cum Len (m)
15 jts 5 7/8" HWDP	5 7/8	4.250	140.00	172.25
X/O	8	2.810	0.79	32.25
3 jts 8 1/4" DC	8 1/4	2.810	28.01	31.46
12 1/8" Stabilizer (COR-3057-AD)	12 1/8	2.810	2.18	3.45
Bit sub with float valve inside	9 1/2		0.95	1.27
12 1/4" Insert Bit-RR	12 1/4		0.32	0.32

Well Control

No.	Depth (mKB)	Stroke (in)	Strokes (spm)	Pres (psi)	Eff (%)	Q (gpm)

Test Date	Last Casing String Run	Test Type	Depth (mKB)	TVD (mKB)	Applied Surface Pressure (psi)
4/8/2014	18 5/8" Shoe, 870.00mKB	Formation Integrity	870.00	843.25	400.0
Applied Surface Pressure (psi)	**Volume Pumped (bbl)**	**Current MAASP**	**Kick Tolerance**	**Dens Fluid (lb/gal)**	**Leak Off Pressure (psi)**
400.0	8.0	400.00		11.48	1,650.4

Geology Information

Formation Name	Prog Top MD (mKB)	Final Top MD (mKB)	Final Top TVD (mKB)
Upper Gadvan		1,209.00	1,156.52
Khalij		1,231.00	1,176.98
Lower Gadvan		1,247.00	1,191.88
Fahliyan		1,286.50	1,228.87
Hith		1,469.30	1,402.67

General Remarks

Comment

>>> Requirements: (60 MT Barite) + (Outstanding of MI MR#10: 120 sxs of Soda Ash) + (MI MR #11) + (150 MT Cement) + (7" Jar) + (12" St Stabilizer).

>>> Sent:
** Borkan to Kish: (HPOGC: 9 5/8" Motor + 12 1/8' Sleeve Stab. + 8" PNMDC + F.Valve + 2 x 9 5/8" 6/7 Lobe Stators + 2 x 8" NMDC + MWD APC) + (9 5/8" x 10 3/4" Al-Ghaith CSG Handling Equipment) + (Aban Equipment) + (7" COR Drilling Jar, s/n: COR3021AD) + (12" COR String Stab) + (Mehran Cementing: 5 x Box of CMT & CMT additives Samples) + (Garbage) + (2 x Jumbo Bags contain CSG Protectors).

>>> Received:
** Mataf pride: (12 1/8" Sleeve Stab.) + (Stationary).